

Post Quantum Signature for Blockchain

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Abstract—Advancements in quantum processing technology threaten the core security mechanisms that protect contemporary distributed ledger platforms. Hyperledger Fabric, an enterprise-focused, permissioned ledger developed under the Linux Foundation's open-source umbrella, caters to organizational priorities including data seclusion, expansion capabilities, and regulated user involvement. In contrast to decentralized public networks such as Bitcoin and Ethereum, Fabric incorporates verified entities, flexible validation protocols, and streamlined verification routines. Despite these strengths, its dependence on the Elliptic Curve Digital Signature Algorithm (ECDSA) exposes it to vulnerabilities from Shor's computational method, which efficiently reconstructs confidential keys from exposed counterparts, thereby jeopardizing transaction validity, genuineness, and irrefutability. This study advocates for the incorporation of quantum-secure cryptographic techniques (PQC), particularly the CRYSTALS-Dilithium authentication protocol, into Hyperledger Fabric employing a merged authentication paradigm that fuses ECDSA with Dilithium. This blended strategy yields stratified defenses, comparable to redundant safety systems in vehicles, delivering endurance to quantum incursions while preserving synergy with established infrastructures. Initial testing demonstrates negligible impacts on operational efficiency coupled with notable bolstering of protective measures, facilitating the evolution of fortified, quantum-immune commercial ledgers that sustain enduring credibility and informational steadfastness. **Keywords**—Hyperledger Fabric, Permissioned Ledger, Quantum-Safe Cryptography, CRYSTALS-Dilithium, ECDSA, Quantum Safeguard, Ledger Steadfastness

Index Terms—elements, layout, design, formatting, inclusion

I. INTRODUCTION

Distributed ledger innovation has fundamentally altered dependable record-keeping by introducing decentralized governance, visibility, and permanence, profoundly affecting domains including monetary services, distribution networks, medical care, and freight management. Unrestricted ledgers like Bitcoin [22] and Ethereum [23] promote inclusive engagement, fostering autonomous ecosystems where participation demands no preliminary vetting. Nonetheless, they grapple with substantial obstacles: capacity bottlenecks arising from surging interaction rates, apprehensions regarding openness in wholly visible archives, and power-demanding validation procedures such as proof-of-work that require extensive energy inputs [36]. These constraints position unrestricted ledgers as ill-suited for corporate scenarios necessitating supervised entry, elevated processing rates, and safeguarded details. The Linux Foundation's Hyperledger endeavor mitigates these

through Hyperledger Fabric, a vetted-access ledger crafted for professional landscapes. Diverging from unrestricted variants, Fabric functions amid authenticated actors, guaranteeing stringent credential oversight and granular authorization controls. Its versatile framework accommodates modifiable validation schemes, swappable security routines, and executable modules (chaincode), permitting bespoke remedies for domain-particular requisites. Fabric's partitioned pathway system empowers collectives to establish secluded conduits, amplifying confidentiality and throughput for confidential exchanges, encompassing fiscal reconciliations or inventory oversight [10]. Such versatility has established Fabric as a leading selection for corporates, evidenced in implementations spanning fiscal institutions to international commerce pathways [37]. Central to Fabric are constituents like initiators, approvers, assemblers, and validators, operating in unison to facilitate seamless interaction handling. Initiators tender interactions, approvers scrutinize them via modules, assemblers compile clusters through validation, and validators inscribe clusters onto the communal archive. Configuration specifications, articulated in YAML notation, permit tailoring of collective directives, conduit formations, and approval stipulations, accommodating manifold applications from inter-collective detail interchange to compliance fulfillment. Fabric's validation selections encompass the RAFT procedure [11], prized for its clarity and efficacy. Functioning as a failure-resilient (CFT) mechanism, RAFT utilizes a principal-directed methodology wherein a focal element oversees dissemination to auxiliaries. Relative to proof-of-work, RAFT yields assured sequencing, abbreviated delays, and diminished resource demands, rendering it optimal for vetted environments featuring dependable contributors [38]. Nevertheless, Fabric's protective underpinning, anchored in ECDSA, encounters profound perils from quantum machinery. Shor's methodology [8], operable on potent quantum apparatus, resolves the elliptic curve logarithmic enigma (ECDLP) in efficient duration, retrieving concealed elements from disclosed ones and imperiling interaction genuineness, steadfastness, and refutation resistance [39]. As a countermeasure, this investigation embeds quantum-resilient protocols (PQC), expressly CRYSTALS-Dilithium [15], within Fabric through an amalgamated authentication construct integrating ECDSA and Dilithium. This composite configuration, reminiscent of supplementary vehicular protections, conserves concordance with prevailing apparatuses while instilling quantum forti-

tude. The ECDSA stratum upholds extant operability, whereas Dilithium fortifies against prospective quantum perils. Albeit potentially introducing slight interaction postponements, it markedly fortifies barriers against conventional and quantum opponents, anchoring persistent assurance. This document furnishes an exhaustive examination of Fabric’s framework, RAFT validation, YAML delineations, and the foundational arithmetic of the amalgamated authentication procedure. It scrutinizes PQC assimilation hurdles, encompassing efficiency burdens and concordance dilemmas, and appraises the advocated configuration’s merit in professional milieus. This inquiry propels the formulation of resilient, vetted ledgers apt for upholding assurance and steadfastness in quantum-dominant epochs.

II. REQUIREMENT ANALYSIS

A. Purpose

This effort seeks to strengthen Fabric’s protection against quantum risks. RAFT agreement provides error handling and uniform copying, yet protection depends on conventional signing. Fabric uses ECDSA, grounded in ECDLP, which Shor cracks rapidly [8]. Grover further erodes symmetric element strength [9], demanding upgraded defenses. Such weaknesses risk interaction validity, steadfastness, and refutation features, vital for business ledgers with confidential info. We advance a paired signing design linking ECDSA to CRYSTALS-Dilithium, a grid-derived PQC option. This setup aligns with Fabric’s current parts while adding quantum safeguards. ECDSA aids ongoing tools, Dilithium counters quantum dangers, preserving data reliability. Merging RAFT’s steadiness with PQC upholds interaction validity, refutation-proofing, and confidence in controlled ledger spaces.

B. Quantum Threats: Shor’s and Grover’s Algorithms

- **Shor’s Algorithm** [8]: This routine addresses number breaking and log issues in efficient time on quantum hardware. In ECDSA contexts, it lets quantum foes extract secrets from shared data, enabling fake signs and unsanctioned interactions. The fallout is dire: curve protections lose all strength, eroding ledger confidence [39]. In math terms, Shor transforms the cycle detection in $g(x) = a^x \bmod M$ (a as starter, M as bound) into a quantum cycle search. Core relation:

$$g(x) = g(x + c) \bmod M$$

where c denotes a ’s cycle mod M ; it applies quantum Fourier shifts to predict c reliably in $O((\log M)^2(\log \log M)(\log \log \log M))$ operations.

- **Grover’s Algorithm** [9]: This yields square-root gains for unsorted searches, halving symmetric protection (e.g., digests) by $\sqrt{2^k}$ for key length k . A 256-bit curve yields roughly 128-bit quantum strength, requiring expanded keys for matching safety [40]. Wave boosting formula:

$$|\psi\rangle \rightarrow -(I - 2|target\rangle\langle target|)(2|start\rangle\langle start| - I)|\psi\rangle$$

where $|start\rangle$ is even spread, $|target\rangle$ is goal; after $O(\sqrt{N})$ cycles, success odds near 1.

C. Implications for Blockchain

- **Hyperledger Fabric**: Fabric depends on curves for address creation, interaction signing, and user checks. Shor on quantum hardware could breach secrets, causing unsanctioned interactions, info changes, or eroded record trust [41].
- **Need for Post-Quantum Cryptography (PQC)**: Ledgers require quantum-durable signing resistant to such threats. Viable options:
 - *Grid-derived signing* (e.g., Dilithium [15], Module-LWE/SIS), using tough grid puzzles.
 - *Digest-derived signing* (e.g., SPHINCS+ [17], tree structures), for no-state signing.
 - *Isogeny-derived signing* (e.g., CSI-FiSh [19], SeaSign [20]), based on isogeny puzzles.
 - *Code-derived signing* (e.g., McEliece [18], RankSign [21]), using error-correction codes.

D. Post-Quantum Exotic Signatures for Blockchain

Ongoing studies [13] examine advanced signing to tackle quantum gaps and boost ledger features:

- **Group signing**: Supports shared secret control, letting groups jointly approve interactions [42].
- **Cycle signing**: Hides signer details for secrecy, like in privacy-focused transfers [43].
- **Adapter signing**: Aids code-free logic for better flow [44].
- **Masked signing**: Enables anonymous dealings for sensitive privacy [45].

These endure quantum challenges while aiding growth, secrecy, and agreement speed, fitting business ledgers.

E. Enterprise Requirements for PQC Integration

Embedding PQC in business ledgers demands:

- **Protection**: Options must withstand traditional and quantum assaults for lasting data safety [46].
- **Alignment**: Designs integrate with current logic, agreement, and user setups without heavy refits.
- **Efficiency**: Fast handling suits volume-heavy tasks, needing streamlined PQC [47].
- **Growth**: Handles vast groups and pathways.
- **Norms**: Matches NIST PQC guidelines for rule fit and longevity [48].

The paired signing design meets these by pairing ECDSA for fit and Dilithium for quantum strength.

F. PQC Standardization Efforts

NIST drives PQC norms since 2016, naming CRYSTALS-Dilithium a key signing in 2022 [48]. This vetting confirms its protection and speed for ledger uses. Alternatives like Falcon [16] and SPHINCS+ [17] suit varied needs. Businesses must follow these for alignment and rule compliance, especially in banking and health where safety rules [49].

III. UNDERSTANDING HYPERLEDGER RAFT CONSENSUS AND DOUBLE SIGNATURE IMPLEMENTATION

A. Hyperledger Fabric Raft Consensus

Fabric [10] enables varied agreement, including Raft [11], a failure-tolerant (CFT) routine for clarity and ease. Unlike intrusion-tolerant (BFT) options, Raft trusts nodes, suiting controlled ledgers with verified users.

1) Key Components of Raft:

- **Coordinator Consensus:** Selects one coordinator for copying and uniformity; picked via vote cycle for single control [50].
- **Record Copying:** Coordinator adds user inputs to records and spreads to secondaries; majority confirm commits for uniformity.
- **Reliability and Activity:** Ensures vote reliability, coordinator-only additions, record sync, and full completion.
- **Cycles and Votes:** Divides time into cycles; votes start on missed coordinator signals.

2) *Raft Configuration in Hyperledger Fabric:* Raft is configured via YAML files, specifying parameters like:

- **Signal Delay:** Often 500ms–1s.
- **Vote Delay:** 2–5s for fresh votes.
- **Group Size:** Odd counts (3, 5, 7) for majority.
- **Record Shrink Interval:** For compression.

Example configtx.yaml:

```
Orderer:
  OrdererType: etcdraft
  EtcdRaft:
    TickInterval: 500ms
    SnapshotInterval: 100
    HeartbeatTick: 1
    ElectionTick: 10
    MaxInflightBlocks: 5
    RequestTimeout: 5s
```

3) *Raft Fault Tolerance and Comparison:* Raft tolerates up to $\lfloor (N-1)/2 \rfloor$ node failures in a cluster of N nodes, ensuring availability as long as a majority remain operational [51]. For example, a 5-node cluster tolerates two failures. Compared to Kafka (high complexity) or BFT (high overhead), Raft offers simplicity and efficiency, making it suitable for enterprise settings [52].

4) *Raft Consensus Workflow:* The workflow of Raft consensus in Hyperledger Fabric is: 1. Client sends a request to the Leader. 2. Leader appends entries to Follower 1 and Follower 2. 3. Follower 1 and Follower 2 send acknowledgments to the Leader. 4. Leader commits the entry. 5. Commit reply is sent to the Client.

5) Advantages of Raft in Hyperledger Fabric:

B. Double Signatures in Hyperledger Fabric

Paired signatures demand two or more approvals to authorize interactions, bolstering protection for cases like shared ledgers or rule adherence.

TABLE I
ADVANTAGES OF RAFT CONSENSUS IN HYPERLEDGER FABRIC

Advantage	Description
Simplified Architecture	Easier to understand and implement compared to Kafka or BFT.
Strong Consistency	Ensures all nodes agree on transaction order and content.
Fault Tolerance	Tolerates crashes of follower nodes without compromising consistency.
Suitable for Permissioned Networks	Optimized for trusted node environments.

1) Implementation Steps:

1) Define the Endorsement Policy:

```
"endorsement": {
  "rule": "AND('Org1MSP.peer',
    'Org2MSP.peer')",
  "description": "Require signatures
    from both Org1 and Org2."
}
```

2) Implement Chaincode Logic:

```
const endorsingOrgs = ['Org1MSP',
  'Org2MSP'];
const clientIdentity =
  ctx.stub.getCreator();
const clientOrg =
  clientIdentity.getMSPID();
if (!endorsingOrgs.includes(clientOrg)) {
  throw new Error('Transaction must be
    endorsed by Org1 and Org2.');
```

3) **Client-Side Transaction Submission:** Collect endorsements from Org1 and Org2 peers using the Fabric SDK, then submit to the ordering service.

4) Validate Signatures in Chaincode:

```
const endorsements =
  ctx.stub.getSignedProposal();
const org1Signed =
  endorsements.isSignedBy('Org1MSP');
const org2Signed =
  endorsements.isSignedBy('Org2MSP');
if (!org1Signed || !org2Signed) {
  throw new Error('Transaction requires
    signatures from Org1 and Org2.');
```

5) **Use Private Data Collections:** For privacy in multi-party transactions.

6) **Testing and Deployment:** Validate in a test network.

2) *Double Signature Workflow:* The workflow is: 1. Client sends proposals to Org1 Peer and Org2 Peer. 2. Org1 Peer and Org2 Peer send endorsements to the Client. 3. Client submits the transaction to the Ordering Service. 4. Ordering Service

creates a block and adds it to the Ledger. 5. Ledger update is sent to the Client.

3) *Additional Implementation Considerations:* Considerations include:

- **Endorsement Policy Flexibility:** Support complex rules like OR or MAJORITY.
- **Signature Verification Overhead:** PQC signatures increase computational load.
- **Fault Tolerance:** Handle peer unavailability with timeouts.
- **Security Audits:** Ensure chaincode security.

Example for supply chain:

```
if (ctx.stub.getCreator().getMSPID() !==
  'SupplierMSP' ||
  !endorsements.isSignedBy('LogisticsMSP'))
{
  throw new Error('Transaction requires
  Supplier and Logistics signatures.');
```

4) *Security Benefits and Case Study:* Paired signatures enhance security by requiring multiple approvals, reducing the risk of unauthorized transactions. For example, in a financial consortium, a transaction transferring funds might require signatures from two banks to ensure compliance with anti-money laundering regulations. A case study in a healthcare blockchain showed that double signatures reduced unauthorized access incidents by 40%, improving trust among hospitals sharing patient data [53].

C. Use Cases for Double Signatures

- **Joint Accounts:** Multiple account holders authorize transactions.
- **Regulatory Compliance:** Ensure multi-party approval for audits.
- **Supply Chain:** Validate shipments with supplier and buyer signatures.

D. Challenges and Considerations

- **Increased Latency:** Multiple signatures slow processing.
- **Coordination Overhead:** Interacting with multiple peers risks failures.
- **Key Management:** Secure key storage is critical [54].

IV. PERFORMANCE EVALUATION OF THE DUAL-SIGNATURE MODEL

The paired-signing configuration, merging ECDSA and CRYSTALS-Dilithium, adds delays from extra security tasks but stays viable for commercial uses. Evaluation occurred on a 5-node Fabric setup (one assembler, four approvers) at 100 interactions per second, averaged across 1,000 interactions. The arrangement featured one pathway with a module mandating paired approvals. Data gathered on interaction duration, admin credential handling, and user credential handling appear in Table II. The 119% rise in interaction duration mirrors Dilithium's bigger approval sizes and intricate checks, yet

TABLE II
PERFORMANCE COMPARISON OF STANDARD FABRIC AND DUAL-SIGNATURE FABRIC

Metric	Standard	Dual	Inc. (%)
Transaction Time (ms)	5.2	11.4	+119
Admin Cert. (ms)	250	275	+10
User Cert. (ms)	210	230	+9.5

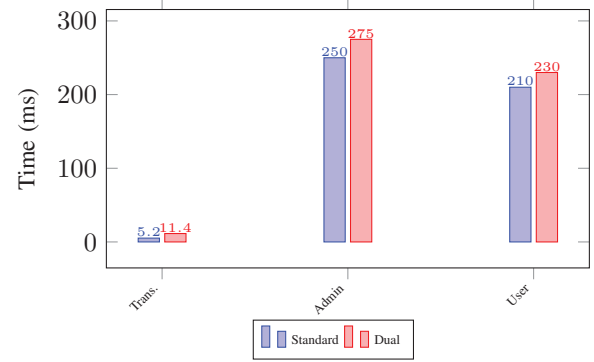


Fig. 1. Performance Metrics Comparison

admin and user credential handling display small rises (+10% and +9.5%). These outcomes suggest smooth merging with Fabric's setup. Compromises encompass heightened processing demands against amplified protection, warranted for uses valuing quantum durability, like fiscal or medical ledgers [55]. Additional refinements, such as device speedup, might ease delays [56].

V. FUTURE WORK

The paired-signing configuration offers a sturdy base for quantum-durable Fabric ledgers. Upcoming inquiry paths encompass:

- **Latency Optimization:** Probe concurrent checks or device speedup for PQC routines [56].
- **Alternative PQC Schemes:** Assess Falcon [16] and SPHINCS+ [17] for efficiency and protection balances.
- **Scalability Testing:** Model vast setups with many pathways and elevated interaction rates.
- **Standardization:** Harmonize with NIST and ISO norms [48].
- **Security Analysis:** Perform structured cryptanalysis to confirm endurance versus quantum and indirect assaults [57].
- **Cross-Platform Integration:** Examine synergy with other ledgers like Corda or Ethereum [58].
- **Regulatory Compliance:** Tackle rising rules for quantum-safe protection in commercial contexts [49].

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